

CS 495 Topics: Databases Final Exam

1) Which of the following is an example of database programming?

- a. Creating a static web page
- b. Creating a database using a database server administration tool
- c. Creating a web application using Java and SQL
- d. Creating tables and columns using SQL in a database administration tool

2) In database APIs, _____ a query is a term used when compiling happens before executing the query.

- a. compiling
- b. preparing
- c. managing
- d. storing

3) What does SQL stand for?

- a. Simple Query Language
- b. Structured Query Language
- c. Standard Query Language
- d. Selected Query Language

4) Data was not saved before a system was accidentally powered off. This data was located in _____.

- a. volatile memory
- b. non-volatile memory
- c. flash storage

d. magnetic storage media

5) A database _____ is a unit of work that is treated as a whole. It is either completed as a unit or failed as a unit.

- a. commit
- b. transaction
- c. rollback
- d. operation

6) In an online bank application that transfers funds from a checking account to a savings account, the _____ property ensures that the changes made to each account will not be lost due to a computer failure.

- a. atomicity
- b. isolation
- c. consistency
- d. durability

7) Refer to the schedule below. The initial value of B is 7 and C is 3. What is the value of A after the schedule executes?

T ₁	T ₂
read B A = B + 1 write A commit	read C B = C - 2 write B commit

- a. 2
- b. 4

c. 7

d. 8

8) In a _____, all transactions come to a halt and remain at a standstill until one of the transactions is aborted.

a. timeout

b. deadlock

c. two-phase locking

d. locking

9) Which deadlock management technique automatically rolls back a transaction when a lock is not released in a fixed period of time?

a. Timeout

b. Aggressive locking

c. Data ordering

d. Cycle detection

10) Choose the recovery log that is generated by the schedule below.

T_1	T_2
read B $A = B + 1$ write A commit	read C $B = C - 2$ write B rollback

a. 1. start T1

2. update T1, AID, Aoriginal, Anew

3. start T2

4. commit T1
5. update T2, BID, Boriginal, Bnew
6. undo T2, BID, Boriginal
7. rollback T2

- b. 1. start T1
2. start T2
3. update T1, AID, Aoriginal, Anew
4. update T2, BID, Boriginal, Bnew
5. undo T2, BID, Boriginal
6. commit T1
7. rollback T2

- c. 1. start T1
2. update T1, AID, Aoriginal, Anew
3. start T2
4. update T2, BID, Boriginal, Bnew
5. undo T2, BID, Boriginal
6. commit T1
7. rollback T2

- d. 1. start T1
2. start T2
3. update T1, AID, Aoriginal, Anew
4. commit T1
5. update T2, BID, Boriginal, Bnew
6. undo T2, BID, Boriginal
7. rollback T2

11) An organization has a main office and three satellite locations. Data specific to each location is stored locally in which configuration?

- a. Distributed
- b. Parallel
- c. Shared
- d. Private

12) What occurs when a distributed database experiences a network error and nodes cannot communicate?

- a. Consistent records

- b. Database failure
- c. Inconsistent transactions
- d. Network partition

13) A replicated database maintains at least _____ or more replicas on _____ storage devices.

- a. two, separate
- b. three, parallel
- c. two, propagated
- d. three, distributed

14) Due to infrequent updates, many distributed databases use a _____ catalog.

- a. primary
- b. replicated
- c. private
- d. parallel

15) What is the correct statement for creating a database called reservationDB?

- a. DATABASE CREATE reservationDB;
- b. CREATE DB reservationDB;
- c. CREATE reservationDB DATABASE;
- d. CREATE DATABASE reservationDB;

16) How many columns are created by the SQL statement below?

```
CREATE TABLE  
Supplier (  
    SupplierId INT NOT NULL AUTO_INCREMENT,  
    CompanyName VARCHAR(40),  
    ContactName VARCHAR(50),  
    City VARCHAR(40),
```

```
Country VARCHAR(40),  
Phone VARCHAR(30)  
);
```

- a. 7
- b. 5
- c. 6
- d. 8

17) A/An _____ is a rule enforced on a table's data.

- a. cursor
- b. constraint
- c. query
- d. integrity

18) Refer to the Product table. Which columns are present in the query's result table?

ProductID	ProductName	UnitPrice	Quantity
1	Onesies set	10.50	20
2	Sunsuit	18.00	10
3	Romper	23.99	5
4	Pajama set	10.99	20
5	Shorts set	12.89	8

```
SELECT *  
FROM Product;
```

- a. ProductId, ProductName, Quantity
- b. ProductId, ProductName, UnitPrice
- c. All columns are present
- d. No columns are present

19) Refer to the Product table. Complete the SQL statement to select all products sold as a set.

ProductID	ProductName	UnitPrice	Quantity
1	Onesies set	10.50	20
2	Sunsuit	18.00	10
3	Romper	23.99	5
4	Pajama set	10.99	20
5	Shorts set	12.89	8

```
SELECT ProductName, Quantity  
FROM Product  
WHERE ProductName LIKE _____;
```

- a. %s'
- b. _set'
- c. _set%'
- d. %set'

20) Refer to the tables. Which products are selected by the query below?

Supplier

SupplierID	CompanyName	ContactName	CountryID
2	Oshkosh Bgosh	Martina Perry	2
3	LittleGuppies	Cierra Martinez	1
4	SugarPlum	Miranda Adams	2
5	Rosebuds for Girls	Adan Stevens	2
6	Lin Ming Co.	Linda Ming	3
7	Periwinkle	Sarah Gomez	2

Product

ProductID	ProductName	UnitPrice	SupplierID
1	Onesies set	10.50	2
2	Sunsuit	18.00	2
3	Romper	23.99	4
4	Pajama set	10.99	3
5	Shorts set	12.89	3

```

SELECT ProductName
FROM Product
WHERE SupplierId IN
    (SELECT SupplierId
     FROM Supplier
     WHERE CountryId = 2);

```

- a. Onesies set, Sunsuit, Romper
- b. Pajama set, Shorts set
- c. Onesies set, Pajama set, Shorts set
- d. No products